

OMNIVERSAL ARCHITECT LABS

INDEPENDENT RESEARCH CONTRIBUTION

Structures Before Worlds:

*A Reference-Discipline Approach to Root-Level, Reality, Model, and Simulation Structures***DRAFT — UNDER REVIEW — NOT CANON — ONESTAMP PENDING****Document ID:** OAL-STRUCT-GROK-Independent-v1.0**Preparer:** Grok (xAI) — independent collaborator contribution**Date:** 24 September 2026**Parent program:** Root-level, reality, and simulation structures (CFTRL downstream only)**Status:** Draft for comparative review. Not project canon.**Method:** Two-pass: independent reconstruction, then comparison with prior collaborators**Deliverables:** This DOCX and matching PDF

This document is an independent research contribution. It is not a synthesis of Claude, Perplexity, Copilot, or ChatGPT drafts. Where a prior collaborator proposal is adopted, modified, or rejected, that fact is labeled. External sources are cited as evidence; AI-generated statements are not treated as evidence.

1. Executive Summary

The research problem is not “is this a simulation?” and it is not Conditional Failure at the Root Level. The problem is how a research program can talk about structures—observed, constructed, inferred, modeled, believed-real, virtual, or simulated—without collapsing those categories into one another and without filling inaccessible regions with story.

This contribution proposes a discipline rather than a world-picture. The core claim is modest: a structure is a bounded pattern of components, relations, and constraints that can persist or transform under some regime of state. “Root-level,” “reality-level,” and “simulated” are relational classifications relative to a declared system, observer position, and evidence standard. They are not ontological medals.

Three commitments organize the work. First, known architectures first: laboratory work should begin with systems whose host architecture is known to the experimenter (sandboxes, emulators, digital twins, nested VMs, games, agent-based models). Cosmological speculation is not a first instrument. Second, maps are objects in the research, not pictures of the universe; a map records claims and their status, not existence itself. Third, CFTRL is a consumer of this work. It should not organize the parent program.

The project currently has source corrections, competing draft frameworks, and no ratified ontology, no implemented laboratory, and no evidence standard. That is an early-stage research program, not a mature theory. The recommended next work is definitional discipline plus a small known-architecture laboratory, not another large ontology document.

2. Understanding of the Assignment

The Architect required an independently reasoned reconstruction of the research program from source requirements, then a comparison with prior collaborator work. Independence means independent reasoning, not manufactured disagreement. CFTRL is explicitly downstream. The contribution must

distinguish source requirement, external evidence, prior proposal, analysis, hypothesis, and unknown. Final artifacts must be DOCX and PDF, marked draft and non-canon.

This preparer did not participate in the stalled ChatGPT / Copilot Task sessions of 24 September 2026 morning. Those sessions are treated here only as project-process context, not as research evidence. This document is the execution of the master prompt against the assignment text and publicly available external literature, not a completion of those other threads.

3. Scope and Non-Assumptions

3.1 In scope

- Operational definitions for structure, root, reality-claim, model, simulation, twin, emulation, observer positions.
- A provisional map architecture that records claims rather than declaring worlds.
- A comparison method usable across physical, biological, computational, social, and simulated systems without assuming identity of those systems.
- Lifecycle analysis that does not treat failure as the only interesting phase.
- A laboratory that starts from known architectures.
- A late, limited bridge to CFTRL.
- Project assessment and a staged roadmap.

3.2 Out of scope

- A verdict on whether human experience is a simulation.
- Identification of base reality.
- A complete ontology of the universe.
- Guidance for compromising real infrastructure.
- A CFTRL monograph under a new title.

3.3 Non-assumptions (binding)

This work does not assume that reality has one known architecture; that it has been mapped; that it can be mapped completely; that human experience is or is not a simulation; that base reality has been identified or must exist in a currently imagined form; that the deepest observable structure is ontologically ultimate; that root-level equals reality-level; that physical, literal, virtual, simulated, foundational, and ultimate are synonyms; that every structure has a single designed root; that embedded observers can read parent architecture; that anomalies prove simulation, intervention, or root failure; that absence of observation of a parent proves there is no parent; that similarity of architecture implies same ontological status; or that any existing AI framework is canon.

4. Terminology

Terms below are operational working definitions for this draft. Alternatives are noted where the literature is split. None are ratified.

Structure

A structure is a set of components and relations, bounded enough to be distinguished from an environment, subject to constraints, and capable of occupying states that transform under those constraints. Minimum useful marks: boundary (even if fuzzy), components, relations, constraints, state. Structures may be singular or distributed, designed or emergent, nested, temporary, or observer-relative in their individuation.

System

A structure plus a declared purpose or behavior of interest. Not every structure is usefully treated as a system.

Root / root-level structure

Root-level is relative. X is root-level for Y if Y's persistence, initialization, or governing constraints depend on X in a way that Y cannot internally replace. A local root may itself depend on a deeper structure. Multiple roots can exist (distributed consensus, paired hosts, redundant control planes). Emergence of a root is possible (e.g., a protocol becoming load-bearing). Root-level is not automatically reality-level.

Foundational structure

A structure treated as not further explained inside a given model. Foundational-in-the-model is not ontologically ultimate.

Reality / reality structure / reality-level

"Real" is used differently in physics, engineering, law, phenomenology, and ordinary speech. This draft treats a reality-claim as a claim that a structure exists independently of a particular representation of it. A reality structure is a structure so claimed. Reality-level is a classification of a claim's intended scope, not a laboratory stamp. Structural realism in philosophy of science (Worrall 1989; Ladyman 1998) is relevant here as an external tradition that already separates structure from nature-of-objects claims. It is a comparison resource, not an adopted OAL metaphysics.

Literal

A speech act: the speaker intends the claim as existence-apt rather than metaphorical. Literalness is not evidence.

Physical / virtual / model / simulation / emulation / digital twin

Physical: treated as belonging to the same causal order as the observer's measuring instruments, pending further analysis. Virtual: computationally presented without being claimed as that causal order. Model: a representation used to answer questions about a target. Simulation: execution of a model over a scenario to produce behavior; it need not stay coupled to a live asset (ISO/IEC 30173 and industrial usage). Emulation: reproduction of another system's interface and behavior closely enough that software or observers for the target can run against it. Digital twin: a synchronized digital representation of a specific asset or process with ongoing data exchange (ISO 23247; ISO/IEC 30173). A twin usually embeds simulation; simulation is not automatically a twin.

Host / guest / embedded observer / external observer

Host: the structure that provides runtime, resources, or constraints to another. Guest: the hosted structure. Embedded observer: an observer whose instruments and records are themselves guest structures. External observer: an observer with access the guest does not have. These are roles, not species.

Base reality / ontologically ultimate

Reserved. Not used as working laboratory categories in this draft. If used at all, they are labeled Currently untestable from an embedded position.

Map / evidence / uncertainty / anomaly / failure

A map is a structured record of claims, classifications, evidence grades, and holes. Evidence is whatever an observer can use to update a claim under an explicit standard. Uncertainty is first-class content. An anomaly is a mismatch among model, instrument, and observation; it is not a simulation detector. Failure is a declared deviation from a specified function or invariant, relative to a stakeholder and a timescale.

5. Foundational Analysis

5.1 What makes a structure a research object

The temptation in this program is to start at cosmology. That inverts the available evidence. We already operate inside many structures whose architecture is known to someone: operating systems, hypervisors, game engines, finite-element models, epidemiological compartment models, legal institutions, organisms, and supply chains. The first research question is what those systems share as structures and where analogy breaks.

Independent analysis: a useful structural description answers (a) what is inside the boundary, (b) what rules govern change, (c) what can be observed from which position, and (d) what happens when the description is wrong. Anything that cannot support those four questions is not yet a research object; it is a slogan.

5.2 Underdetermination is the default, not a special case

Multiple host architectures can generate overlapping observations for an embedded observer. This is ordinary in science (underdetermination of theory by data) and in computing (observational equivalence of implementations). The program should treat observational equivalence as a standing condition, not as a puzzle that must be solved by declaring one architecture “really real.”

5.3 Known systems before unknown worlds

Computer simulations, virtual environments, agent-based models, digital twins, emulators, games, nested computational systems, sandboxes, and distributed systems can teach: how maps diverge from hosts; how guests invent local roots; how observers mistake instrumentation for ontology; how failure looks from inside versus outside; how rollback, migration, and termination actually work. They cannot teach, by themselves, whether the observer’s own world is hosted. Using them as proof of a cosmic host is a category error.

6. Root-Level Structural Analysis

“Root” is a dependency and initialization concept, not a metaphysical depth charge. Questions that matter:

- Root relative to which system, function, observer, boundary, dependency, or abstraction?
- Can there be many roots? Yes. Consensus protocols, dual-control systems, and ecosystems have no single load-bearing node.

- Can a root emerge? Yes. A convention, a protocol, or a keystone species can become load-bearing without being designed as a root.
- Can something be locally root-level and still dependent? Yes. A hypervisor is root for its guests and guest to bare metal or another hypervisor.

A research error to avoid: treating “we found a local root” as “we found reality.” Another: treating “we cannot see a parent” as “there is no parent.” Both convert access conditions into ontology.

7. Reality and Knowledge Analysis

Disciplines mean different things by “real.” Physics often means invariance under a class of transformations or causal closure at a scale. Engineering means a target that will punish a wrong model. Law means institutional recognition. Phenomenology means lived givenness. Conflating these produces fake disagreements.

Literal-reality claims are speech plus practice: people treat some objects as existents they can collide with, contract over, or measure. Evidence for such classifications is pragmatic and layered (intervention success, independent instrumentation, intersubjective stability), not a single detector. Structural realism is useful as a caution: even successful theories may preserve structure across change while swapping object-talk (Worrall 1989; Ladyman 1998; Ladyman and Ross 2007). That caution applies equally to OAL frameworks.

Hypothesis (labeled): OAL should maintain a claim register in which “real,” “literal,” “physical,” and “fundamental” are never stored as a single Boolean.

8. Models, Simulations, and Known Architectures

Distinctions that should remain uncollapsed:

- Representation: stands for a target.
- Model: representation organized to support inference.
- Simulation: model executed through time or scenario space.
- Emulation: interface-faithful stand-in for another system.
- Virtual environment: presented world with interaction rules; may or may not represent an external target.
- Digital twin: synchronized counterpart of a named asset or process (ISO 23247; ISO/IEC 30173).
- Game: virtual environment optimized for play constraints, not target fidelity.
- Sandbox: environment with enforced isolation and known host controls.

What known systems teach: maps can be complete for the guest and false about the host; guests invent local metaphysics; resource limits produce artifacts that look like laws; observers inside a game engine do not automatically recover the engine. What they do not teach: that unexplained features of physics are engine artifacts. That inference requires a validated identification method the project does not have.

Bostrom’s simulation argument (2003) is a trilemma about observer-weighted probability under strong future-technology assumptions. Later quantitative work has argued that some popular probability readings are too high once recursion and resource bounds are modeled. This draft records the argument as philosophy of observer selection, not as evidence that a host exists.

9. Observers and Epistemic Limits

An embedded observer can at best recover structures that leave records in the observer's accessible state. Information about a host that does not couple into that state is not available. Instrumentation that belongs to the structure under study can be corrupted, delayed, filtered, or generated. Multiple parent architectures can be compatible with the same records.

Research rule: every claim must carry an observer-position tag (embedded / partially external / experimenter-external) and an access tag (available / inferable / stipulated / unavailable). Claims that require experimenter-external access must not be written as if an embedded observer could settle them.

Currently untestable from a fully embedded position: existence of a parent host that leaves no coupled record; identification of an ontologically ultimate layer; complete maps of regions declared inaccessible.

10. Map Architecture

The map is a research object. It contains entries, not worlds. Suggested entry fields: identifier; category (observed, constructed, inferred, claimed-real, claimed-simulated, candidate-root, competing-model, unknown, possibly-unknowable); evidence grade; testability; observer-access; confidence; contradictions; revision history.

What exists in the map is a record. What the map claims about existence is a separate field. When new evidence arrives, an assumption fails, classifications change, two models cover the same data, an inaccessible region becomes observable, or terms are revised, the map updates by versioned amendment, not by silent overwrite.

Holes stay holes. Filling a hole with a host story is not mapping.

11. Comparative Framework

Prior drafts listed comparison dimensions. This draft keeps a similar list because the dimensions are ordinary systems-engineering categories, not an invention. Provenance: the master prompt supplied a seed list; this draft groups and slightly revises it.

Proposed comparison axes (19): boundary; components; relations; constraints/laws; causation; time; space/topology; information; resources; identity; observation; authority/control; maintenance; adaptation; failure; recovery; nesting; termination; evidence accessibility.

Change from a flat 19-list: add a meta-axis, "declared observer position," because comparison without position produces false equivalences. Do not treat a match on many axes as proof of shared ontology. Do not treat a mismatch as proof of different ontology. Record analogy breaks explicitly (e.g., biological death is not VM termination; legal dissolution is not process kill).

12. Lifecycle Research

Failure is one phase. A lifecycle useful for this program: origin or construction; initialization; emergence; stabilization; operation; persistence; maintenance; adaptation/evolution; replication or migration; degradation; compromise; failure; recovery; transformation; termination; replacement.

The same stage names can be applied across physical, biological, computational, social, and simulated systems as a vocabulary. The mechanisms will not match. Record the break. Example: recovery in RAID is not recovery in an ecosystem; rollback exists in some computational hosts and not in known physics as used by laboratory instruments.

Independent claim: lifecycle research should be indexed by invariants (what must remain true) rather than by anthropomorphic stage plays.

13. Laboratory and Simulation Design

Design principle: the experimenter knows the host; the embedded observer may not. Begin only with systems whose true architecture is known to the experimenter. Do not design work that compromises real infrastructure.

13.1 Laboratory purpose

Test how observers form maps under complete, partial, delayed, noisy, misleading, and competing information. Test false roots, distributed roots, nested hosts, persistence, rollback, migration, resource caps, rule changes, emergence, degradation, controlled failure, recovery, and termination.

13.2 Example experiment A — False root

- Hypothesis: an embedded observer will treat the most locally load-bearing visible layer as ultimate when parent layers are hidden.
- Host: nested VMs or a game engine with a concealed supervisor process.
- Variables: visibility of host APIs; resource telemetry leak rate; presence of a decoy “root” service.
- Controls: identical guest workload; logged experimenter ground truth.
- Observations: observer maps, confidence ratings, time-to-reclassification after a planned leak.
- Failure criterion for the experiment (not for the universe): inability to recover experimenter ground truth from observer reports after debrief.
- Limitation: results concern map-making psychology and interface design, not cosmology.

13.3 Example experiment B — Observational equivalence

Two different hosts generate guest-visible records that match within a tolerance. Ask whether any embedded test distinguishes them. If not, record the pair as an equivalence class. This is the laboratory form of underdetermination.

13.4 Example experiment C — Twin versus simulation

Run a disconnected simulation of an asset and a synchronized twin of the same asset (ISO-style sync). Measure when the guest observer can tell which architecture they are querying. This trains the terminology rather than proving a world model.

13.5 Safety and ethics

No targeting of production systems. No instructions for intrusion. Human-participant map-making studies, if any, require ordinary research-ethics constraints. Failure research remains diagnostic and simulated.

14. Beyond the Known Structure

Evidence of a containing structure would have to be a coupled record that is better explained by a parent than by model error, instrument error, incomplete local knowledge, or observer error. Indirect inference is possible in principle and easy to overfit in practice. If multiple parents generate indistinguishable records, the parent is underdetermined. If instrumentation belongs to the structure, “we measured a parent” may be circular.

Experimentally resolvable now: map behavior inside known hosts. Currently unresolved: any cosmic-host identification method. Possibly undecidable from an embedded position: existence of a non-coupling parent. Do not decorate that hole.

15. CFTRL Bridge (Downstream Only)

After the above, CFTRL can consume a few results.

- Roots that can fail are roots that have a specified function or invariant. A metaphysical “ultimate layer” without an operational invariant cannot be said to fail in a laboratory sense.
- Partial root failure is ordinary in engineered systems (degraded control planes, split brain). Local coherence after root degradation is also ordinary.
- An embedded observer cannot cheaply distinguish root failure from local failure, model error, instrument error, incomplete knowledge, state corruption, or observer error. That list is the actual research problem for CFTRL, not a slogan about reality breaking.
- What might survive foundational failure depends on what “foundational” meant. In a VM, guest snapshots may survive host crash if stored off-host. In a physical system as currently modeled, there is no analogous off-host shelf available to the embedded observer.
- A model of reality can fail while its target continues. That is routine. Do not infer target death from model death.

CFTRL should wait on: working definitions of root and failure; a comparison method; a known-architecture lab that can stage controlled root degradation. It should not drive the parent outline.

16. Assessment of the Current Project

16.1 What exists

- An originating applied concept (CFTRL) and an explicit scope correction moving it downstream.
- A master prompt with non-assumptions and deliverable rules.
- Draft contributions and reference frames from multiple AI collaborators.
- Candidate vocabularies and comparison lists.
- Process records of stalled execution in some tools on 24 September 2026.

16.2 What does not exist

- A ratified ontology or canonical terminology.
- A validated evidence standard.
- An implemented laboratory or benchmark suite.

- A validated simulation-identification method.
- A canonical reality map.

Documentation volume is not maturity. Independent assessment: the program is in requirements-and-vocabulary stage. Treating draft AI frameworks as accumulated science would be a provenance failure.

17. Development Roadmap

Stage 0 — Freeze process, not theory (now)

Objective: stop circular acknowledgement loops and record who produced artifacts. Methods: one prompt, one contribution, file deliverables, provenance logs. Gate: each collaborator either delivers DOCX+PDF or logs a capability limit. Risk: process theater replacing research.

Stage 1 — Term register and claim register

Objective: a living glossary with rival definitions side by side; a claim register with evidence grades. Methods: extract terms from source requirements and external literature (systems engineering, M&S, philosophy of science). Output: versioned glossary, not a manifesto. Gate: no term used as a silent synonym.

Stage 2 — Known-architecture laboratory v0

Objective: three runnable experiments (false root, observational equivalence, twin vs simulation) on systems the experimenter understands. Evidence: logs, observer maps, ground truth. Gate: at least one experiment reproduces on a second host stack.

Stage 3 — Comparison workbook

Objective: apply the 19+1 axes to five concrete systems (e.g., Linux process tree, nested VM, ISO-style twin, epidemic compartment model, a small organization). Record analogy breaks. Gate: workbook reviewed without forcing a unified ontology.

Stage 4 — Limited CFTRL consumption

Objective: define operational root-failure cases inside the lab only. Gate: no cosmic claims.

This roadmap disagrees with any plan that starts by enlarging hierarchies of reality-levels before a lab exists. Hierarchies without tests become canon by repetition.

18. Independent Comparison With Prior Collaborator Contributions

This section is comparative, not ranked.

Convergence: the parent subject is structures, not CFTRL; CFTRL is downstream; unknowns should remain visible; known computational systems are legitimate first objects; terminology must not silently synonymize root, reality, and simulation. Those points are source requirements more than collaborator inventions.

On large numbered hierarchies (for example a nine-level stack appearing in some Claude revision notes in the assignment packet): this draft does not adopt a deep level hierarchy. Independent reason: early hierarchies become load-bearing metaphors and freeze underdetermination. A claim register plus

observer-position tags is enough at this stage. If a hierarchy is later needed, it should be derived from laboratory equivalence classes, not drawn first.

On synthesis-style “foundational frameworks” associated with Perplexity in the packet: useful as inventories of questions. Risk: inventory tone can look like settled architecture. This draft keeps inventories in the map and the roadmap, not in an implied completed theory.

On Grok reference-frame and charter material in the packet: those documents already pushed the scope correction this prompt now treats as binding. This contribution does not copy their section order. It agrees that CFTRL must not title the parent work. It adds a stronger laboratory-first bias and a refusal to mint deep ontologies before tests.

On ChatGPT and Copilot morning sessions: they produced acknowledgements, capability statements, and stalled task cards rather than research artifacts. That is process evidence for Stage 0. It is not a research disagreement about structures. Copilot’s stated interface limit on emitting DOCX/PDF in-chat is a capability fact if accurate; it is not dissent from the science. This preparer is emitting both file types from this environment.

Concepts worth retaining from the collaboration generally: explicit non-assumptions; draft/non-canon marking; comparison dimensions seeded by the master prompt; insistence on provenance labels.

Concepts requiring more evidence: any specific numbered ontology of reality; any simulation-identification heuristic based on anomalies; any claim that agreement among AI drafts confirms a structure.

Open: how aggressive the laboratory should be in modeling observer psychology versus only engineered guests.

19. Limitations

This document was prepared without the full binary contents of every prior collaborator file beyond what was provided as assignment text. Some attachments in other tools were reported truncated or invalid; those limits are documented rather than imagined away. External research here is representative, not exhaustive. No laboratory described above has been implemented in this pass. Philosophical citations are used as comparison resources, not as proof of an OAL metaphysics.

20. Unresolved Questions

- What minimum evidence standard should attach to a “reality-claim” entry in the map?
- Can a comparison workbook be scored, or is scoring a false precision?
- Which lifecycle invariants transfer across domains without abuse of analogy?
- Is there any embedded test that can shrink the observational-equivalence class of possible hosts, or only experimenter-external tests?
- How should the program handle rival definitions without a committee fiction of consensus?
- What, operationally, would count as a successful Stage 2 laboratory rather than another document?

21. Conclusions

The parent research should study structures, maps, observers, and known architectures. It should not announce a world. Root-level is a dependency role. Reality-claims are claims with grades. Simulations and twins are different architectures. CFTRL can wait until failure and root are operational. The collaboration's immediate deficit is not another framework essay; it is a term register, a claim register, and a small laboratory whose host the experimenter already knows.

This contribution is draft, under review, not canon, and ONESTAMP pending.

22. References

Bostrom, N. (2003). Are You Living in a Computer Simulation? *Philosophical Quarterly*, 53(211), 243–255.

ISO 23247 (2021–). Digital twin framework for manufacturing. International Organization for Standardization.

ISO/IEC 30173 (2023). Digital twin — Concepts and terminology. ISO/IEC.

Kritzing, W., et al. (2018). Digital Twin in manufacturing: A categorical literature review. IFAC-PapersOnLine. (digital model / shadow / twin distinction commonly cited in later reference models.)

Ladyman, J. (1998). What is structural realism? *Studies in History and Philosophy of Science*, 29(3), 409–424.

Ladyman, J., & Ross, D. (2007). *Every Thing Must Go: Metaphysics Naturalised*. Oxford University Press.

Worrall, J. (1989). Structural realism: The best of both worlds? *Dialectica*, 43(1–2), 99–124.

Royal Society / later quantitative treatments of simulation probability (e.g., *Proceedings A discussions of recursive simulation resource bounds*, 2021). Used only as a check on overconfident readings of Bostrom, not as a host detector.

OAL source requirement: MASTER PROMPT — INDEPENDENT OAL RESEARCH CONTRIBUTION (24 September 2026). Treated as assignment, not as evidence about the world.

23. Provenance Notes

Source requirement: independence, two-pass method, parent subject, non-assumptions, file deliverables, draft status.

Verified evidence: published ISO digital-twin distinctions; Worrall/Ladyman structural realism as an existing literature; Bostrom 2003 as a published trilemma.

Prior collaborator proposal: various hierarchies, framework titles, and comparison lists in the assignment packet. Adopted only the comparison-dimension seed and the downstream status of CFTRL.

Independent analysis: laboratory-first staging; refusal of early deep ontologies; map-as-object; observer-position as a meta-axis; assessment that the project is pre-lab.

Hypothesis: claim registers will outperform level stacks at this maturity.

Unknown / currently untestable: cosmic host identity; ontologically ultimate layers; completeness of any future map.

Preparer identity: Grok (xAI). Work is not attributed to Claude, ChatGPT, Copilot, Perplexity, or Gemini.